CabinGuard-ADI: Bounded-Evidence Multimodal Detection of Acute Driver Incapacitation with an Authenticated Minimum-Risk Response

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***Abstract*—Driver monitoring systems detect drowsiness and distraction, not the sudden loss of driving capacity caused by cardiac syncope or a convulsive seizure. CabinGuard-ADI separates these two events from normal driving using four contactless channels: camera pulse rhythm, eyelid and head pose, a headrest accelerometer, and seat and steering-wheel sensors. The evidence of every channel is bounded so that no single sensor can trigger a vehicle response; a minimum-risk manoeuvre needs two physical sensors that agree for 2 seconds. The pulse-rhythm branch is trained on public PhysioNet recordings, including 22.2 hours of road driving, ten annotated seizures and ventricular arrhythmia onsets; the motion branch only on recorded motion, including the neck accelerometry of clinical seizures in 115 patients; both are evaluated across held-out records, subjects and patients. On out-of-fold composite episodes the fused detector started no manoeuvre in 22.2 hours of real driving, a 95 % upper bound of 0.14 per hour, against 3.92 false starts per hour for the pulse branch alone, and detected 79/80 syncope episodes and 8/10 seizure episodes built from clinical seizure motion, with median latencies of 12.0 and 22.2 seconds from onset. A gateway accepts only authenticated, fresh, plausible CAN commands and completes the stop if the detector fails. A signed and encrypted 76-byte medical extension reaches the emergency centre, while nearby vehicles are warned without the diagnosis. The eye, head and seat channels still use modelled likelihoods, and alone the motion branch raises alarms during real driving; the Phase 2 bench measures both.**

***Index Terms*—Driver monitoring, driver incapacitation, syncope, epileptic seizure, multimodal fusion, remote photoplethysmography, minimum risk manoeuvre, SecOC, eCall**

I. Introduction and Motivation

Sudden illness of the driver is a small but persistent cause of fatal crashes: in Finland and in the Swiss canton of Vaud it caused 1.5 % and 3.4 % of all traffic deaths, and probable cardiac arrest alone about 2 % of driver deaths [1]. Since July 2024 new European vehicles must carry drowsiness warning and new types also distraction warning, systems that the same regulation says should work without biometric information [2]. Neither function addresses a driver who loses consciousness or motor control within seconds.

Two events dominate that scenario. Cardiac syncope is a transient loss of consciousness from cerebral hypoperfusion caused by an arrhythmia [3]; the driver becomes flaccid, the head drops and the hands leave the wheel. A convulsive seizure begins with a tonic phase of sustained contraction and continues with clonic, rhythmic jerking [4]. The two look different only on average: myoclonic jerks occurred in 90 % of syncopes induced in healthy volunteers [5], so a motion sensor alone confuses them, while a camera alone cannot tell closed eyes from a blinded lens. In a driving simulator, every seizure with motor involvement or impaired consciousness impaired driving [6], so a detector must also catch seizures that do not move the body.

The contributions are a fusion rule under which no single sensor can start a manoeuvre, with the bound derived from the decision threshold; pulse-rhythm and motion detectors evaluated on public real recordings, including real road driving; a gateway that accepts only authenticated, plausible commands and fails safe; and an emergency message that gives the etiology to the emergency centre alone.

II. Related Work

Camera-based pulse measurement projects the skin colour signal onto a plane orthogonal to the skin tone (POS) [7]; in a moving car, narrow-band near-infrared video reduces the effect of changing daylight [8]. Real-road driver ECG comes from the driver-stress study of Healey and Picard [9], and eye closure is commonly measured with the eye aspect ratio [10].

Seizure detection outside hospitals relies mostly on wearables: a wrist accelerometer detected 35 of 39 generalized tonic-clonic seizures with 0.2 false alarms per day [11], and an under-mattress movement sensor detected 16 of 23 convulsive seizures and none of 197 non-convulsive ones [12]. Clonic discharges last about 0.2 s with silent periods of exponentially increasing duration [13], and heart rate rises during most seizures [14]; none of these studies measures motion through a vehicle seat.

On the vehicle side, UN Regulation No. 157 defines a minimum risk manoeuvre that an automated lane keeping system performs when a transition demand is not answered [15]. In-vehicle networks without authentication let a compromised node command brakes [16], cameras can be blinded with commodity lasers [17], and AUTOSAR secure onboard communication (SecOC) adds truncated message authentication codes and freshness values to CAN frames [18].

III. Problem Definition

Every 100 ms the system assigns the driver to a class $C \in \{\mathrm{N}, \mathrm{S}, \mathrm{Z}\}$ for normal driving, cardiac syncope and con-

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vulsive seizure, and decides whether to start a minimum risk manoeuvre (MRM). It must meet five requirements: R1, few false MRM starts per hour of normal driving; R2, detection within seconds of onset; R3, no single sensor able to start an MRM; R4, safe behaviour under sensor, processing and communication failure and under tampered or unauthorized messages; R5, only the minimum necessary data leaving the vehicle. A false start first produces a 3 s alert that a conscious driver cancels, so R1 counts alerts as well as braking.

IV. System Architecture

The sensing layer has three physical sensors, shown on the left of Fig. 1. A 30 fps camera with 850 nm illumination on the steering column yields the pulse beat times, the eye aspect ratio e and the head pitch θ. An MPU6050 accelerometer in the headrest is sampled at 100 Hz. A force-sensing resistor in the seatback, a second one in the wheel rim, two capacitive touch pads and the steering torque τ form the seat and wheel sensor. An ESP32 node streams them in 24-byte frames with sequence number and CRC to the edge host, which runs detection, the watchdog and the MRM state machine and commands the vehicle only through one authenticated CAN message to the gateway that owns the actuators. Inputs older than 300 ms count as missing.

V. Detection Methodology

A. Features

The camera front end detects pulse peaks in the POS signal [7]; in the public recordings used for training the beats are ECG R-peaks instead. Over a 10 s window the pulse-rhythm vector $\mathbf{c}$ holds the rate from the median inter-beat interval, its difference from the driver's own running baseline, the coefficient of variation of the intervals, and the fraction a of the window not covered by intervals shorter than 2 s. Ventricular fibrillation, flutter and asystole produce no pulse and raise a towards one. Eye opening is the eye aspect ratio of Soukupová and Čech [10] from six eyelid landmarks.

$$e = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\,\|p_1 - p_4\|} \tag{1}$$

In Equation (1), p_1 and p_4 are the eye corners and p_2, p_3, p_5, p_6 the upper and lower lid points in image coordinates. Clonic motion is characterised by the share of acceleration power in the 2–6 Hz band, computed over a 2 s window from the sum $S(f)$ of the three per-axis Welch spectra [19].

$$r = \frac{\int_2^6 S(f)\,\mathrm{d}f}{\int_{0.5}^{20} S(f)\,\mathrm{d}f} \tag{2}$$

In Equation (2), r is the spectral energy ratio and f the frequency in hertz; each window is first low-passed at 7 Hz by a causal fourth-order Butterworth filter, so recordings sampled between 16 and 100 Hz present the same bandwidth. A ratio alone is noisy when the body is still, so the motion vector $\mathbf{m}$ adds the 2–6 Hz RMS acceleration, the peak frequency, the total RMS, the largest autocorrelation at lags of 0.15–0.5 s, which captures the repetition of jerks, and the peak-to-mean ratio of the band spectrum. Postural slump ψ is one minus the mean normalized seatback pressure over 1 s, and g is the grip state: active, disengaged, or clenched when the rim force exceeds a threshold.

B. Branch Evidence

Four branches turn features into evidence. The cardiac branch is a multinomial logistic regression on $\mathbf{c}$ and the motion branch gradient-boosted trees [20] with the histogram splits of LightGBM [21] on $\mathbf{m}$, both trained with weights that balance the classes. The vision branch on (e, θ) and the posture branch on (ψ, g) use expert-set Gaussian and categorical class densities, since no public recording provides them for incapacitated drivers. A discriminative model's posterior becomes a likelihood ratio once its training prior is divided out, the scaled-likelihood step of hybrid connectionist recognisers [22].

$$\lambda_b(C) = \ln \frac{P_b(C \mid \mathbf{x}_b)}{\pi_b(C)} - \ln \frac{P_b(\mathrm{N} \mid \mathbf{x}_b)}{\pi_b(\mathrm{N})} \tag{3}$$

In Equation (3), $\lambda_b(C)$ is the log-likelihood ratio of class C against normal driving from branch b, $\mathbf{x}_b$ the branch input and π_b the prior the model was trained under, uniform here because of the balanced weights. The motion branch only separates clonic motion from everything else, so it gives $\lambda_{\mathrm{mot}}(\mathrm{S}) = 0$.

C. Bounded-Evidence Fusion

Each branch's evidence for an event is limited to a bound c by shifting both event classes together, which keeps the branch's preference between syncope and seizure.

$$\tilde{\lambda}_b(C) = \max\Big(-c,\ \lambda_b(C) - \max\big(0, \max_{C' \neq \mathrm{N}} \lambda_b(C') - c\big)\Big) \tag{4}$$

In Equation (4), $\tilde{\lambda}_b(C)$ is the bounded evidence for $C \neq \mathrm{N}$ and C' ranges over the two event classes. Treating the branches as conditionally independent given the class, the product rule for combining classifiers [23] gives the fused posterior.

$$P(C \mid \mathbf{x}) \propto \pi(C) \exp\Big(\sum_{b \in \mathcal{A}} \bar{\lambda}_b(C)\Big) \tag{5}$$

In Equation (5), $\mathcal{A}$ is the set of branches whose sensors are available, $\bar{\lambda}_b$ the average of $\tilde{\lambda}_b$ over the last $W = 2$ s, and π the operational prior with $\pi(\mathrm{N}) = 0.8$ and 0.1 for each event. With the decision threshold $\Gamma = 0.85$ and the prior log-odds $l_0 = \ln(0.1/0.8)$, the bound is chosen so that one branch cannot reach Γ while two agreeing branches can.

$$l_0 + c < \ln \frac{\Gamma}{1 - \Gamma} < l_0 + 2c \tag{6}$$

Equation (6) holds for $1.91 < c < 3.81$, and $c = 3$ is used. A missing branch simply leaves the sum in Equation (5), so sensor loss never needs a separate model.

D. Decision Rule

The watchdog sets the flag F that starts the MRM when one event class keeps $P(C \mid \mathbf{x}) \geq \Gamma$ for $T_{\mathrm{ver}} = 2$ s and at least two physical sensors give $\bar{\lambda}_b(C) \geq 1$. The camera feeds two branches but counts as one sensor, so a blinded lens reporting closed eyes and no pulse cannot decide alone.

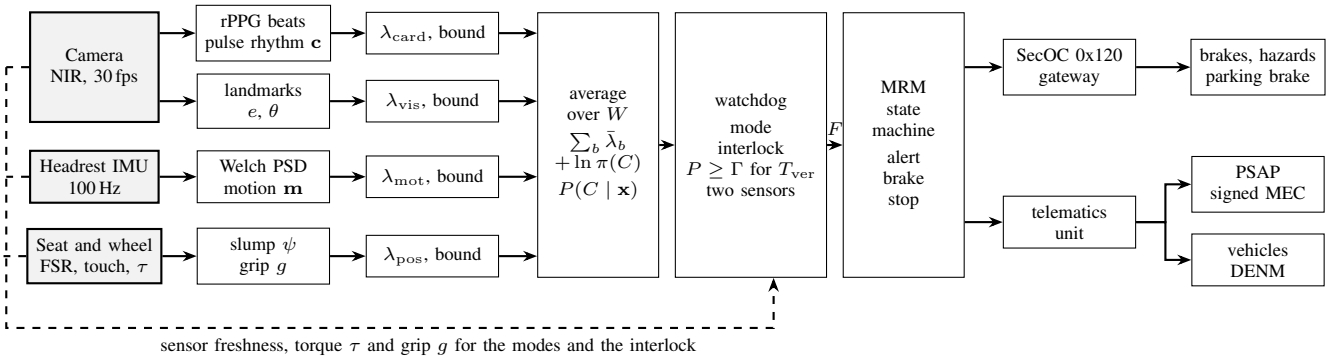


Fig. 1. CabinGuard-ADI signal flow from the three sensors to the vehicle actuators and the two emergency receivers.

VI. Safety Response and Fault Handling

Once F is set, the state machine follows the transition-demand sequence of UN Regulation No. 157 [15] with design values of our own: a 3 s audible and haptic alert that the driver cancels by braking or by more than 4 Nm of steering torque, unless the torque coincides with clonic motion or a clenched grip; hazard lights and escalation; braking at $-3.2\,\mathrm{m/s^2}$ from 10 s after the alert with a move to the shoulder; and at standstill the parking brake and door unlock. The emergency call is placed when the alert goes unanswered, not after the stop.

The watchdog derives the mode from sensor freshness. Losing the camera removes the cardiac and vision branches (degraded mode 1), losing the accelerometer removes the motion branch (degraded mode 2), and with fewer than two sensors left no MRM may start and the driver is warned instead (degraded mode 3). When the camera reports no pulse or closed eyes while the wheel shows an upright driver steering with more than 1 Nm, the readings contradict each other, so the camera is distrusted for 10 s and the event is logged.

The gateway trusts no command. A frame must pass authentication and a plausibility gate that allows only the phase order of the state machine and decelerations between 0 and $-4\,\mathrm{m/s^2}$; it does not check how long each phase lasts, so a compromised host holding the key could brake at $-4\,\mathrm{m/s^2}$ within three frames. The command doubles as an alive signal: after 300 ms without a valid frame the gateway flags an ECU fault and, if the driver is still in control, actuates nothing; if a manoeuvre is already under way it completes the stop, because a driver judged incapacitated cannot take over. The host itself goes silent after three consecutive 100 ms deadline overruns. The emergency message is retried with 1, 2 and 4 s backoff, then sent by SMS, then kept and retried every 30 s; the manoeuvre never waits for it.

VII. Cybersecurity and Threat Model

The protected assets are the vehicle actuators, the driver's physiological data, the integrity of the emergency message and the driver's identity and location. The sensor link inside the cabin is protected against corruption by its CRC but not against an attacker with physical access; that attacker and a compromised host holding the CAN key are outside the scope, a residual risk; the CAN bus, the cellular network and the radio channel are untrusted, and the gateway is the trusted boundary in front of the actuators. Table I lists the threats and the control that answers each.

TABLE I
Threats, controls and the fault-matrix case that exercises each

Threat	Control	Test case
Forged command	24-bit AES-CMAC, SecOC profile 1	unauthorized key
Replayed command	64-bit freshness counter, 8 bits sent	replayed frames
Tampered frame	MAC check, alive timeout	tampered frames
Valid MAC, unsafe value	phase order, $-4\,\mathrm{m/s^2}$ limit	implausible command
Camera blinding or spoofing	cross-sensor interlock, two-sensor rule	optical blinding
Sensor loss or stale data	freshness, degraded modes	IMU, seat stale
ECU hang	fail-silent host, gateway stop	two hang cases
Network outage	retry, SMS fallback, queue	two bearer cases
Message forgery, eavesdropping	ECDSA P-256 signature, ECIES to PSAP	unit tests

The 0x120 command follows SecOC profile 1, AES-128-CMAC [24] truncated to 24 bits over the data identifier, payload and a 64-bit freshness value of which the low 8 bits are sent [18]. Its payload, the CAN database file and the decoder are tested to agree bit for bit. The status frame that other nodes can read carries sensor health only, never a biometric value.

The 76-byte medical extension container (MEC) has a 12-byte header with etiology, confidence, latency, heart rate and sequence number, and a 64-byte ECDSA P-256 signature from a pseudonymous key, the curve and signature used by ITS security certificates [25]. It is encrypted to the emergency centre's public key with ephemeral ECDH, HKDF and AES-GCM and travels beside the eCall, not inside its 140-byte minimum set of data [26], which already carries the location. Nearby vehicles receive a decentralized environmental notification with cause code 93, human problem, and sub-cause 0, unavailable [27], so the etiology never leaves the emergency channel. The system stores no video and no waveform; only

TABLE II
PUBLIC RECORDINGS USED FOR TRAINING AND EVALUATION

Database	Content	Role
Driver stress [9]	ECG, 17 drives	normal, 22.2 h
Partial epilepsy [29]	ECG, 10 seizures	seizure, normal
Arrhythmia [30]–[32]	ECG, 105 records	syncope, normal
Walk, climb, drive [33]	hip, wrist, 32 adults	normal, 10.4 h
SeizeIT2 [34]	neck, 115 patients	seizure, normal
Seizure mimics [35]	wrist, 6 adults	seizure, normal
MHEALTH [36]	chest, 10 adults	normal
UCI HAR [37]	waist, 30 adults	test only

10 s feature buffers exist, in memory.

VIII. EVALUATION METHODOLOGY

Table II lists the public recordings, all from PhysioNet [28]. Driving-study R-peaks come from the WFDB XQRS detector. Windows in VT are labelled cardiac and keep their beats; windows in fibrillation, flutter or asystole are labelled cardiac and lose their beats, because no pulse reaches the face. Atrial fibrillation, bigeminy and other rhythms that do not incapacitate count as normal, as hard negatives.

The cardiac branch is validated leaving one of 130 records out. The motion branch is trained on recorded motion only: motor seizures recorded with a neck accelerometer during video-EEG monitoring [34] and volunteers' seizure mimics [35], against patient background more than 10 min from any seizure, the volunteers' other activities, driving, walking and stairs [33], and everyday activities [36]. Seizure windows are positives only when their motion exceeds twice the pre-ictal level and the median hip RMS of the training drivers, as weaker motion is lost in road vibration; this labelling took three held-out runs, the earlier ones having taught the branch to flag stillness. Sample weights balance the classes and sources. People are held out in five folds; the mimics are scored on their test split, which shares its six volunteers with training; [37] is never trained on. No recording measures seizure motion in a seat, so a jerk train with the clonic dynamics of [13], 0.2 s discharges with silent periods growing from 0.05–0.15 s to about 1 s over 40 jerks, is added to held-out hip driving at 0.05–0.5 g, for testing only.

The fused detector is evaluated on composite episodes, which is the conditional-independence assumption of Equation (5) applied to the test data. A normal episode pairs a held-out drive with held-out driving accelerometry. A syncope episode takes one of 80 ventricular arrhythmia onsets after 30 s of normal rhythm; loss of consciousness, driving the vision and posture channels, is assumed 6–12 s later. A seizure episode takes each of the ten seizures and the neck motion of a motor seizure of a held-out patient from its onset, or, after a 5–15 s tonic phase, the modelled jerk train at 0.1, 0.2 and 0.5 g. Vision and posture channels are generated with everyday confounders: blinks, 1–3 s glances down every 45 s, forward leans every 120 s and one-handed steering every 60 s. Separately, the full software pipeline runs 14 failure and attack cases with synthetic sensors, 10 seeds each.

TABLE III
BRANCH RESULTS UNDER HELD-OUT RECORDS AND HELD-OUT SUBJECTS

Cardiac branch	Gaussian NB	Logistic	Boosted trees
AUC, VT/VF vs rest	0.928	0.955	0.936
AUC, seizure vs rest	0.825	0.846	0.790
Seizure records hit, of 7	7	5	6
Motion branch		**Held out**	
Convulsive seizures detected		42/52	
Hyperkinetic; tonic or clonic		40/270; 4/29	
Automatisms; non-motor		6/67; 10/283	
Alarms/h, patient background		6.59	
Alarms/h, driving at hip; wrist		9.7; 6.8	
Mimics; other recordings fired		26/34; 6/104	

TABLE IV
FUSED DETECTOR ON OUT-OF-FOLD COMPOSITE EPISODES

Branches	False/h	Syncope	Seizure, clin.; mod.	Latency, s
Cardiac only	3.92	47/80	5/10; 15/30	7.5, 30.5
Motion only	0.00	0/80	0/10; 0/30	–, –
Vision only	0.00	80/80	10/10; 30/30	12.0, 13.0
Posture only	1.13	79/80	10/10; 30/30	14.0, 5.2
Cardiac, motion	0.09	0/80	1/10; 16/30	–, 110.5
All four	0.00	79/80	8/10; 29/30	12.0, 22.2
Camera lost	0.09	0/80	2/10; 28/30	–, 57.5
IMU lost	0.14	79/80	10/10; 28/30	12.0, 15.0

IX. RESULTS

Table III gives the branch results. The pulse rhythm separates ventricular arrhythmia well and seizures less well, as a clear heart-rate change accompanies only about 70 % of partial-onset seizures [14]. The fixed rule $r > 0.65$ fires 34.3 times per hour of real driving. On held-out patients the motion branch detected 42/52 convulsive seizures from 36 patients but few other seizures, which a neck sensor barely registers. Alone it raised 9.7 alarms per hour of held-out driving at the hip and 6.8 at the wrist. Without the everyday activities in training, these rose to 12.4 and 9.5, and cycling and running raised 29 and 30 alarms instead of 1 and 4, for 41/52 convulsive seizures detected.

The deployed cardiac model is the logistic one, the best by AUC. In Table IV only the cardiac and motion rows use real signals; vision and posture are generated from the densities that score them, so every row including them is optimistic. The cardiac branch alone raises 3.92 false starts per hour; adding the motion branch cuts this to 0.09 but cannot confirm syncope. All four branches started no manoeuvre in 22.2 hours of real driving, which puts the 95 % upper bound of the rate at 0.14 per hour, and detected 79/80 syncope and 8/10 clinical-motion seizure episodes, with 90th-percentile latencies of 14.6 and 39.4 s, 29/30 with the modelled jerk train and 10/10 with recorded mimics. Without the IMU 10/10 clinical-motion episodes are detected: reading weak seizure motion as normal, the motion branch votes against camera and seat. Losing the camera confirms no syncope, by design.

The window W was searched over 0, 2, 4 and 8 s, with the median syncope latency growing from 11.5 to 16.5 s. In each

of 5 episode folds, W was chosen on the other folds as the shortest window with the highest syncope sensitivity; every fold chose 2 s, so the held-out results are those of Table IV.

All 140 of 140 fault-matrix runs reached the required outcome: no manoeuvre under sensor loss, stale data, blinding, or tampered, replayed, forged or implausible commands; a fail-silent gateway when the host hangs while driving and a completed stop when it hangs during a manoeuvre; and a verified MEC after cellular loss or a 60 s loss of every bearer. On one laptop core a detection cycle took 5.8 ms at the median, 11.0 ms at the 99th percentile and 81 ms at worst, within the 100 ms deadline.

X. Conclusion

Bounded evidence and two physical sensors made the detector sensitive and quiet on real driving, and the vehicle moves only on commands a fail-safe gateway authenticates. Three limits remain. The vision and posture likelihoods are expert-set and their channels modelled, so the fused results above are an upper estimate until those channels are measured; the motion branch learns seizure motion from patients lying in bed with a neck sensor, raises alarms during real driving on its own, and no recording yet measures seizure motion at a headrest; and the composite episodes assume that the branches are independent given the class. The code already runs on an ESP32 sensor node, a camera and a CAN bus, where Phase 2 will measure the eye, head and seat likelihoods, headrest seizure motion and pulse accuracy.

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